

Tutorial – Week 11

Black-box Testing

Question 1

Consider a software application that calculates the monthly payments of a car loan. The application accepts three inputs:

- Cost of the car: The cost should be a positive integer between \$1,000 and \$200,000.
 - Loan period: The period of the loan in years should be a positive integer between 1 and 10 years.
 - Interest rate: The annual interest rate should be a positive decimal between 0.01 (1%) and 0.2 (20%).
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- a. Create Normal Boundary Value cases to test the functionality of this application.
 - b. Create Robust Boundary Value cases to test the functionality of this application.
 - c. Create Worst-case Boundary Value cases to test the functionality of this application.
 - d. Create Robust Worst-case Boundary Value cases to test the functionality of this application.

Question 2

Consider a function $f(x1, x2)$ where the values of $x1$ and $x2$ are defined to be $1000 \leq x1 \leq 10000$ and $10\text{-year} \leq x2 \leq 50\text{-year}$.

Assume the equivalence classes for $x1$ and $x2$ are: -

V1 = [1000 to 5000]

V2 = [5001 to 10000]

V3 = [10-year to 19-year]

V4 = [20-year to 29-year]

V5 = [30-year to 50-year]

- a. Using graphical representation, show the minimum test cases one could generate using weak normal equivalence class testing.
- b. Why a strong normal equivalence class testing generates more test cases compare to a weak normal equivalence class testing?

- c. List the equivalence classes of x1 and x2 for robustness testing.
- d. Using graphical representation, show the minimum test cases you could generate using the weak robust equivalence class testing.

Question 3

Consider an online e-commerce platform that calculates shipping costs based on three factors: the type of item being shipped (Fragile or Non-Fragile), the chosen shipping method (Standard, Express, or Overnight), and the total order price (Below \$50, \$50-\$100, Above \$100).

The rules for shipping costs are as follows:

- Standard shipping costs \$5 for Non-Fragile items and \$10 for Fragile items.
- Express shipping costs \$10 for Non-Fragile items and \$15 for Fragile items.
- Overnight shipping costs \$20 for Non-Fragile items and \$25 for Fragile items.
- Orders below \$50 incur an additional handling fee of \$5, regardless of item type or shipping method.
- Orders between \$50 and \$100 have no additional fees.
- Orders above \$100 get free shipping, regardless of item type or shipping method.

Create a decision table to represent the logic of this system and describe how you would convert this decision table into test cases for the e-commerce platform.